

*Question 1:* We use the following model to describe the short-term transmission dynamics of an outbreak of measles in one region:

$$\frac{dS}{dt} = -\frac{\beta IS}{N}, \quad \frac{dE}{dt} = \frac{\beta IS}{N} - \sigma E, \quad \frac{dI}{dt} = \sigma E - \gamma I, \quad \frac{dR}{dt} = (1 - p_D)\gamma I, \quad \frac{dD}{dt} = p_D\gamma I \quad (1)$$

with contact rate,  $\beta = 5 \text{ days}^{-1}$ , a mean latency period of 10 days, and a mean infectious duration of 3 days. It follows the recovery rate,  $\gamma = 1/3$  and the rate at which exposed individuals become infected,  $\sigma = 1/10$ . We assume the population size is 100,000 and at the start of the outbreak, 90% of the population have immunity.  $D$  denotes the class representing deaths that occur due to measles,  $p_D = 0.01$  denotes the probability of death due to measles for an infected person and  $S, E, I, R$  have the standard interpretation.

*Q1a):* Assuming no prior immunity,  $R_0 = \beta/\gamma = 15$ . Furthermore, if we assume a prior immunity of 90%, then we have

$$R_e(0) = \frac{\beta S(0)}{N\gamma} = \frac{\beta \cdot 0.1 \cdot N}{N\gamma} = \frac{0.1 \cdot \beta}{\gamma} = R_0 = 1.5.$$

*Q1b)*

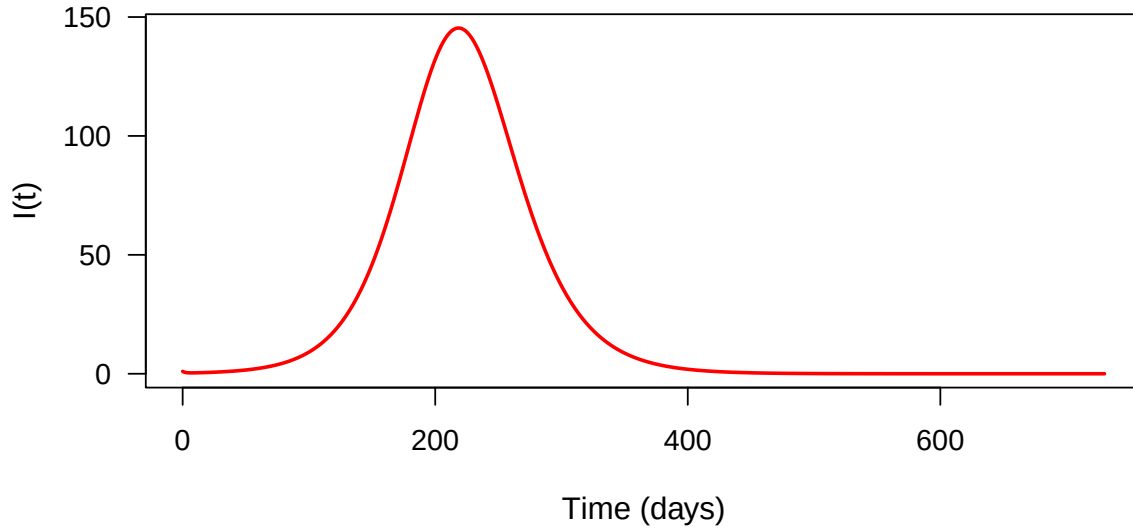


Figure 1: Infection class dynamics over 2 years.

It suffices to run `Deaths <- ceil(tail(Classes$D, n = 1))` since the deaths compartment has positive rate. Therefore, we have 59 deaths over the 2 year period.

*Q2a)*

| Events           | Rate                 | Change                                 |
|------------------|----------------------|----------------------------------------|
| 1. New Infection | $\frac{\beta SI}{N}$ | $S \mapsto S - 1$<br>$E \mapsto E + 1$ |
| 2. End Latency   | $\sigma E$           | $E \mapsto E - 1$<br>$I \mapsto I + 1$ |
| 3. Recovery      | $\gamma I$           | $I \mapsto I - 1$<br>$R \mapsto R + 1$ |
| 4. New Death     | $p_D\gamma I$        | $R \mapsto R - 1$<br>$D \mapsto D + 1$ |

Table 1: Table of events with rate and changes for analogous stochastic model.

We note that whilst another possibility is to split the I compartment into the R and D compartments together, we take everything into the R compartment then with a probability of  $p_D$  take elements into the D compartment. Both achieve the same overall effect. *Q2b)*

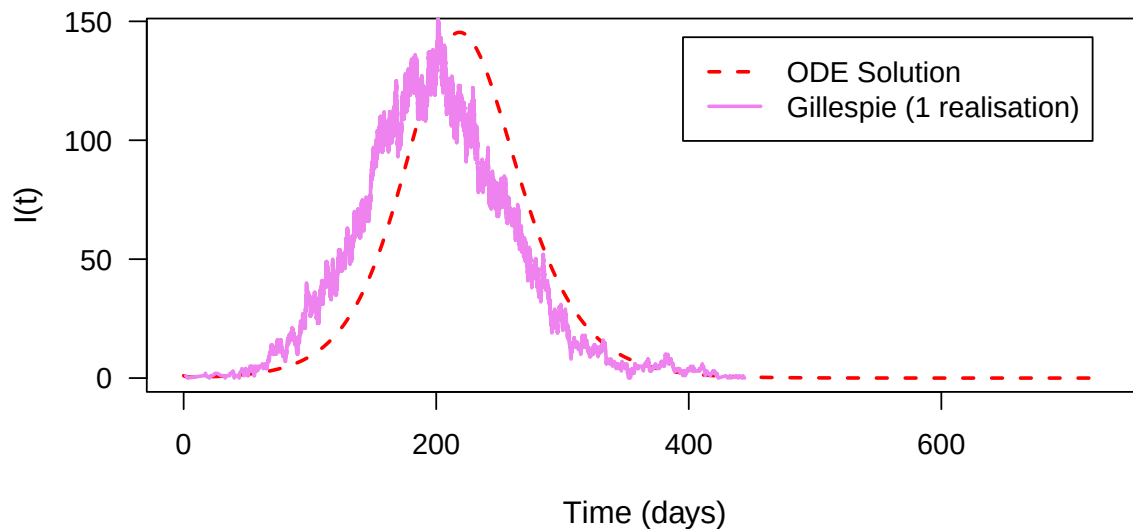


Figure 2: Infection class dynamics over 2 years, comparing ODE to Gillespie 1 realisation.

*Q2c)* For our Gillespie realisation, using the R code `round(Classes.G$t[LastInft])` where `LastInft <- tail(which(Classes.G$I == 0), n = 1)`, we obtain that the duration of the epidemic was 444 days. By taking the last entry of the recoveries vector, we further obtain that the final size was 95,746.

*Q2d)*

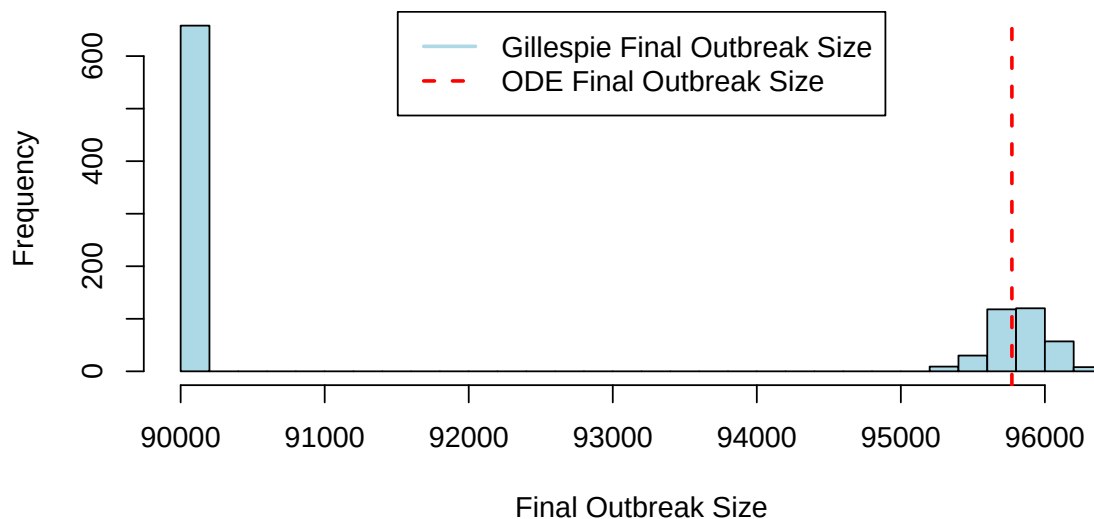


Figure 3: Distribution of final outbreak sizes over 1000 Gillespie realisations compared to ODE final outbreak size.

It took us 19.08 seconds to run 1000 realisations of our Gillespie algorithm to compute a distribution of final outbreak sizes. The large bar at 90000 is due to the simulation immediately doing the *death* event causing the single infected person to enter the death compartment and end the simulation.

*Q2e)* In our setup, since 90% are assumed to have immunity, the rate of infection is given by  $\frac{\beta \cdot 0.1 \cdot N}{N} = 0.1\beta$ . Since there is initially a single infection, the rate of recovery

is  $\gamma$  and the rate of death is  $\gamma \cdot p_D$ . Given an infected individual can either recover or die, we have

$$\mathbb{P}(\text{infect before death or recovery}) = \frac{0.1 \cdot \beta}{0.1 \cdot \beta + \gamma + \gamma \cdot p_D} = \frac{150}{251} =: A.$$

So, it follows

$$\mathbb{P}(\text{death or recovery before infection}) = 1 - A = \frac{101}{251} =: B.$$

Then, with the assistance of branching diagrams, we can consider what the possible situations look like.

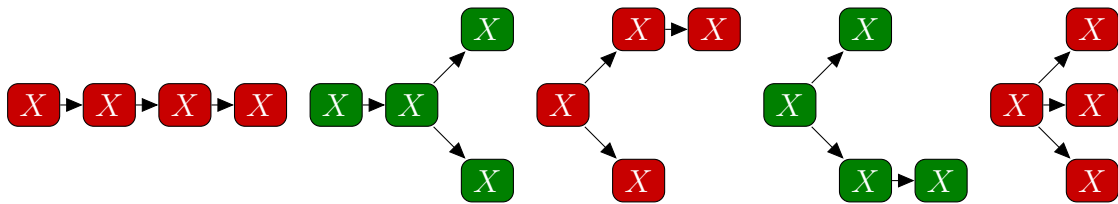


Figure 4: Diagram showing possible configurations for exactly 3 additional infections

Then, the probability of exactly 0, 1, 2 or 3 additional infections with a single initial infection is

$$\mathbb{P}(0, 1, 2, 3 \text{ additional infections}) = B + AB^2 + 2 \cdot A^2B^3 + 5 \cdot A^3B^4 =: C \approx 0.574.$$

Therefore, if we run 1000 realisations of the epidemic, we expect  $1000 \cdot C \approx 574$  (57.4%) outbreaks to have only 0 – 3 additional infections.

*Q2f)* Comparing this result to our model as shown in Figure (3), we observe that near 700 of the realisations went extinct almost immediately which is significantly above our 57.4% derived above. This is because in our calculation we restrict ourselves to up to 3 additional infections but in our simulation this number could be higher. Observe that the number of configurations (an example seen in the above figure) for  $k$  additional infections is given by  $C_k$ , the  $k^{\text{th}}$  Catalan number. Therefore, the probability for up to an additional  $n$  infections and then extinction is

$$\sum_{i=0}^n C_i \cdot A^i \cdot (1 - A)^{i+1} = \sum_{i=0}^n \binom{2i}{i} \cdot \frac{1}{i+1} \cdot A^i \cdot (1 - A)^{i+1} \xrightarrow{n \rightarrow \infty} 0.6733.$$

This probability, 67.33% aligns better with our simulation data in Figure (3), and is the probability that the outbreak will go extinct before creating a large epidemic.

*Q3a)* We consider long-term infection dynamics and elimination by introducing demography into our model. We expand our population to 1,000,000. Assuming the natural per capita death rate is  $\mu = 1/60 \text{ years}^{-1}$ , a proportion  $v$  of infants are vaccinated at birth and this vaccine has  $p_e = 0.95$  probability of success, our model is governed by the

following ODE equations:

$$\begin{aligned}
\frac{dS}{dt} &= (1-v)\mu N + (1-p_e)v\mu N - \frac{\beta IS}{N} - \mu S \\
&= (1-p_e v)\mu N - \frac{\beta IS}{N} - \mu S, \\
\frac{dE}{dt} &= \frac{\beta IS}{N} - \sigma E - \mu E, \\
\frac{dI}{dt} &= \sigma E - \gamma I - \mu I, \\
\frac{dR}{dt} &= \gamma I - p_D \gamma I - \mu R, \\
\frac{dD}{dt} &= p_D \gamma I, \\
\frac{dV}{dt} &= p_e v \mu N - \mu V.
\end{aligned}$$

Here, the  $V$  compartment represents successfully vaccinated new borns. It should be noted we assume the frequency-dependent infection term to have  $N$  (a constant) as a denominator despite death from measles. Then, the term  $(1-v)\mu N$  represents non vaccinated susceptibles meanwhile the term  $(1-p_e)v\mu N$  represents new borns who were unsuccessful when vaccinated. Therefore, upon expansion,  $vp_e\mu N$  represents the successful new borns who were vaccinated. Furthermore, for each compartment,  $A$ , we include a  $-\mu A$  term for natural deaths whilst not including this for the  $S$  class because we replace each death in other compartments with a new susceptible as shown in the table of events below.

| Events                        | Rate                 | Change                                 |
|-------------------------------|----------------------|----------------------------------------|
| 1. Successful vaccination     | $p_e v \mu N$        | $S \mapsto S - 1$<br>$V \mapsto V + 1$ |
| 2. Natural death (S)*         | $\mu S$              | $S \mapsto S$                          |
| 3. Natural death (V)          | $\mu V$              | $V \mapsto V - 1$<br>$S \mapsto S + 1$ |
| 4. New Infection              | $\frac{\beta IS}{N}$ | $S \mapsto S - 1$<br>$E \mapsto E + 1$ |
| 5. End Latency                | $\sigma E$           | $E \mapsto E - 1$<br>$I \mapsto I + 1$ |
| 6. Natural death (E)          | $\mu E$              | $E \mapsto E - 1$<br>$S \mapsto S + 1$ |
| 7. Natural death (I)          | $\mu I$              | $I \mapsto I - 1$<br>$S \mapsto S + 1$ |
| 8. Recovery                   | $\gamma I$           | $I \mapsto I - 1$<br>$R \mapsto R + 1$ |
| 9. New Death (due to measles) | $p_D \gamma I$       | $R \mapsto R - 1$<br>$D \mapsto D + 1$ |
| 10. Natural death (R)         | $\mu R$              | $R \mapsto R - 1$<br>$S \mapsto S + 1$ |

Table 2: New table of events with rate and changes for analogous stochastic model.

\*Since we replace any deaths by a new susceptible, *event 2* is not an event but included for completeness.

Q3b)

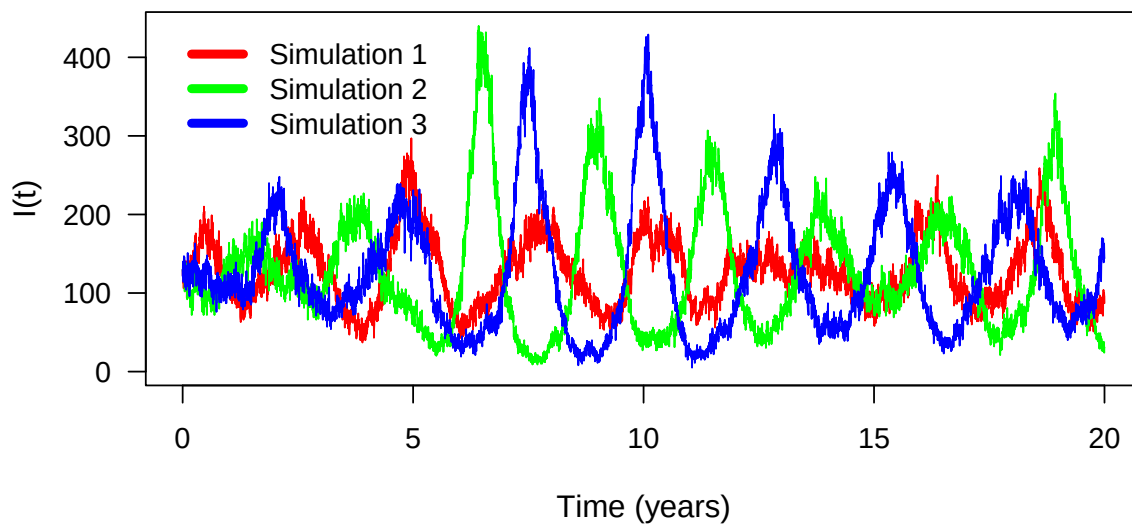


Figure 5: Infection dynamics for three realisations of the SEIRDV model using the tau-leaping stochastic method, across 20 years with no vaccination.

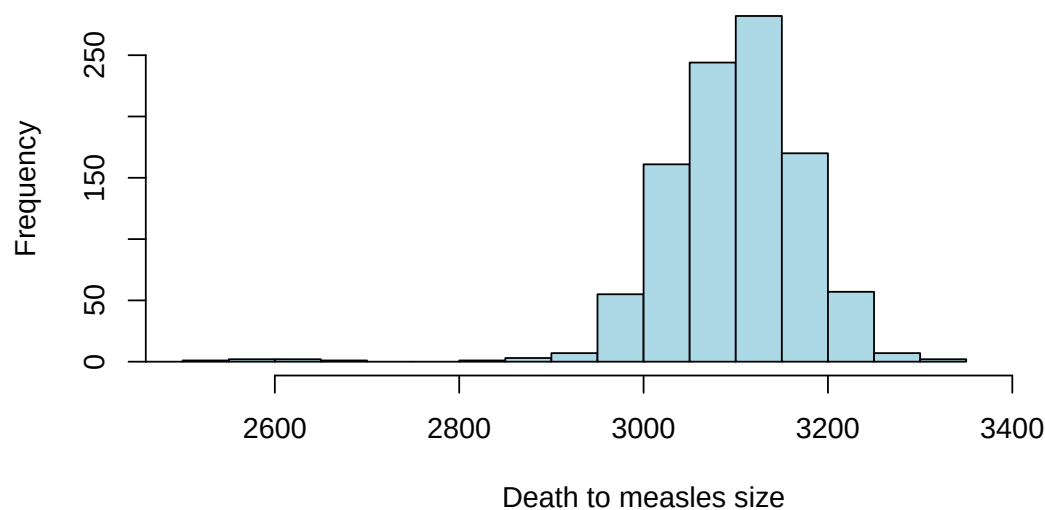


Figure 6: Distribution of deaths due to measles compartment over 1000 realisations for SEIRDV model using tau-leaping stochastic method.

We observe that in Figure (5), there are around 7 peaks for infection prevalence and the general trend seems to be periodic local almost-elimination of measles within our 20 year simulation period, however the probability of elimination is near 0 for this model.

Q3c) Tau-leap has a faster simulation time compared to the Gillespie algorithm because it has less computational cost by performing multiple events every timestep. For this SEIRDV model, we have a large number of events and a much larger simulation time therefore using the Gillespie algorithm with this model would take much longer and use more computational resources compared to the tau-leap algorithm and thus the latter is preferred in this case.

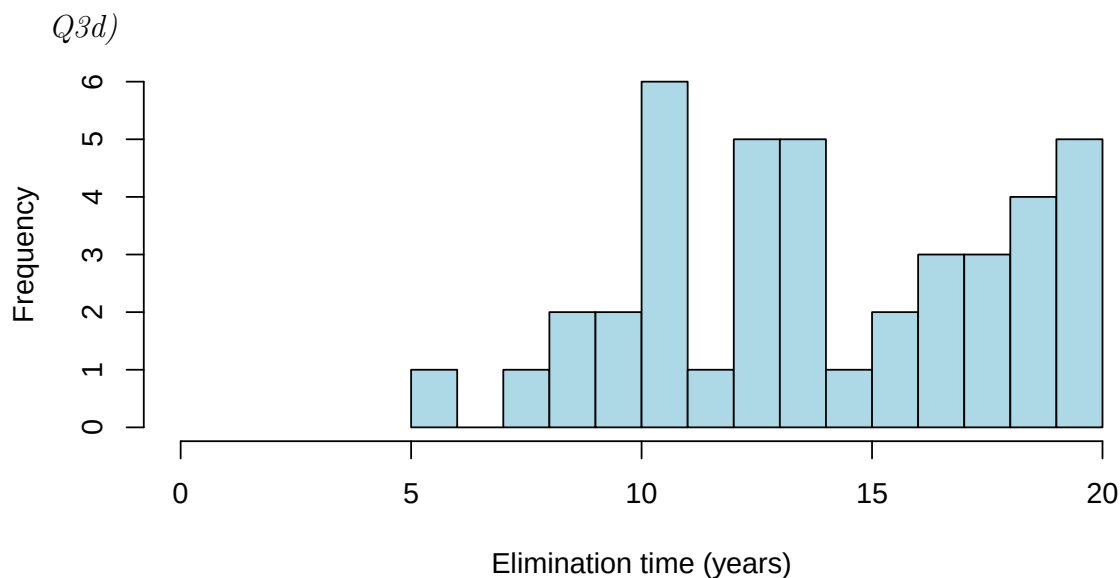


Figure 7: Distribution of elimination times over 1000 realisations for SEIRDV model using adaptive tau-leaping stochastic method with no vaccination.

To determine elimination time, we run a check for when both the  $I$  class and  $E$  class are zero. Observe that not all 1000 realisations are included in the plot above because the majority of simulations didn't reach 0 infections within the 20 year duration. The simulations assume no vaccination so this is an expected result. While our tau-leaping method took 201.08 seconds to run 1000 realisations, Cao's adaptive tau leaping algorithm took 822.70 seconds to run 1000 realisations. Two key reasons this package is desirable is

1. Cao's adaptive tau-leaping algorithm produces more accurate simulations by using smaller timesteps when the model has a lot of events going on at once and larger timesteps when the simulation has less happening.
2. Adaptive methods have better control over the error in approximation by adjusting the timestep whenever the models error exceeds the specified  $\varepsilon$  threshold.

One reason why this package may not be desirable is

1. Additional computational overhead, adaptive tau-leaping requires additional computations to determine the optimal time step,  $\tau$ . When our model has a consistent rate through the whole duration of the simulation then we can assume a fixed timestep and use regular tau-leaping without the need to adjust timestep.

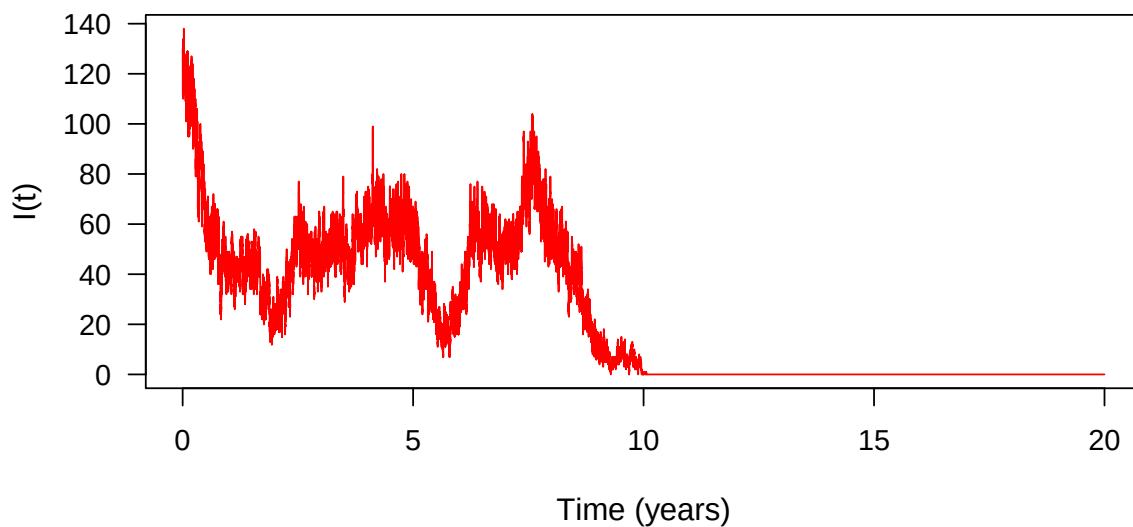
$Q^{3e})$ 

Figure 8: Infection dynamics for one realisation of the SEIRDV model using Cao's adaptive tau-leaping method, across 20 years with proportion vaccinated set to 60%.

Comparing this to Figure (5), we observe that the infection is completely eliminated after 10 years where as before we had periodic oscillations. This is because previously we had 0% of people being vaccinated and therefore with new borns being susceptible the cycle of measles would continue. By introducing vaccinations, we can keep infection prevalence to 0 and maintain it with vaccination. To compute the probability that this vaccination coverage leads to local elimination within 5 years, we plot a histogram of elimination times as we did in Figure (7) and determine what proportion of realisations have an elimination time of at most 5 years. As seen in the R code, we compute this to be 0.293.

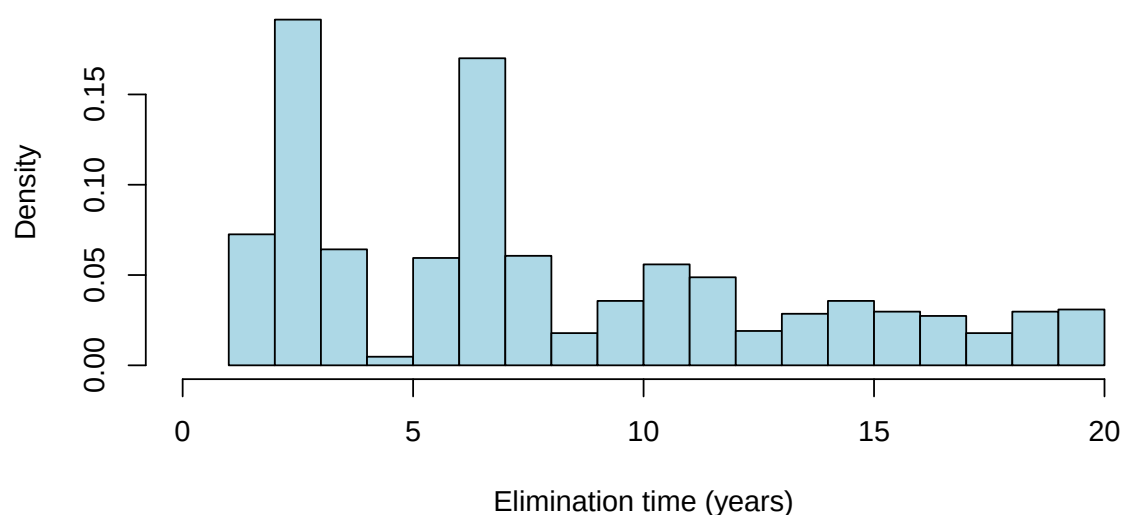


Figure 9: Distribution of elimination times over 1000 realisations for SEIRDV model using adaptive tau-leaping stochastic method with 60% vaccination coverage.

Now, we try to use a binary search approach to identify a value for  $v$  such that the probability (based on a Monte Carlo simulation) of elimination within 5 years exceeds 0.95.

After running the simulation 1000 times using our Cao's adaptive tau-leaping method for  $v = 0.78$  we yield a probability of 0.933 and for  $v = 0.79$  we yield a probability of 0.942 and after repeating the experiment we get 0.951 suggesting that a vaccine coverage of at least 79% is needed so that the probability of elimination within 5 years is more than 95%. For  $v = 0.8$  we yield a probability of 0.961 so to be safe we say  $v = 0.8$  is our answer to this question.

*Q3f*) In the video, it discusses that the DRC has recurrent measles outbreaks due to a low vaccination coverage. This is evident in our model where in Figure (5) we see multiple outbreaks due to there being no vaccination. In Q3e, we find the coverage required to eliminate measles within 5 years, however, in many rural areas of the DRC, weak healthcare infrastructure, conflict, and logistical challenges disallow them to achieve the desired coverage. In both the video and article, the main challenge is lack of income to buy the vaccinations. Poor road networks and limited cold chain storage make vaccine transportation difficult. Additionally, in both sources, it discusses ongoing armed conflicts and insecurity in certain areas which puts healthcare workers at risk and hinder access to vulnerable populations. Despite these efforts, the MSF have experience deploying mobile vaccination teams, using innovative cold chain solutions, and collaborating with local leaders to improve vaccine acceptance. If they are able to increase coverage to around 79%, as we saw in our model in Q3e, then infection prevalence of measles could be better contained and controller in rural DRC. The MSF aim to improve feasibility of providing vaccination in rural areas of the DRC by using innovative cold chain solutions such as portable, solar paneled refrigerators for storing the vaccine which partially solved the transporation problem and making access to these facilities free which addresses the financial problem.